Econometrics R

R code functions and output interpretation · W1, W2, W5, W6 · 52 questions

1. Which symbols assign a value to a variable in R, and what does an assignment do?

```
x <- 5
y = x + 2
```

2. Which function combines several values into one vector in R, and how does it work?

```
try1 <- c(9, 8, 7, 6)
```

3. What happens when a single number is added to a vector (try1 + 5), and what happens when two vectors of the same length are added?

```
try1 <- c(9, 8, 7, 6)
try2 <- c(5, 4, 3, 2)
try1 + 5
try1 + try2
```

4. In an R data frame, what do the rows and the columns represent?

```
View(df)
dim(df)
```

5. What statistics does summary() report when it is applied to a numeric vector or to a data frame?

```
summary(x)
```

6. When a data set contains extreme outliers, which measure of the center is more appropriate, the mean or the median, and why?

```
mean(x)
median(x)
```

7. Name two statistical features of a data set that you can read from a histogram created with hist().

```
hist(x, breaks = 15)
```

8. How can table() and sort() be combined in R to find the mode of a numeric variable?

```
table(x)
sort(table(x), decreasing = TRUE)
```

9. In name.R, what does read_file("WAGE2_description.txt") combined with string splitting functions accomplish regarding variable names?

10. In the W2 dataset description, lwage is defined as the natural log of wage. What is the primary econometric purpose of using lwage instead of raw wage?

11. In the W2 dataset, variables like married, black, south, and urban are coded as binary indicators (e.g., =1 if married). How are these interpreted in regression?

12. In Lab 2_2, what is the precise statistical function of setting na.strings="." when importing data via read.csv()?

13. What are the functional role and the potential risk of using attach(practice) and detach(practice) in R?

14. When computing sum(competence - mean(competence)), why does the resulting value theoretically equal zero?

15. Why do we take the square root of variance to compute standard deviation instead of using variance directly?

16. When data is positively skewed with outliers, why is the Interquartile Range (IQR) preferred over standard deviation?

17. In Lab 3_2, what is the key benefit of using formula notation (e.g., mean(mpg ~ am, data = mtcars)) from the mosaic package?

18. Why would an analyst use the explicit package syntax base::mean(mpg ~ am, data = mtcars)?

19. In Lab 4_2, what is the functional advantage of using sapply(data$openingHour, to12) over a standard loop?

20. What does the stringr function str_detect(str, fixed(",")) return, and how is it used in data pipelines?

21. Read the code. In the formula perform ~ eye, which variable is the dependent variable and which is the independent variable, and how does plot() use them?

```
plot(perform ~ eye)
```

22. The code draws a scatterplot of perform on eye, the OLS line (blue), a horizontal line at the mean of perform (red) and a vertical line at the mean of eye (green). Where does the blue line pass, and what does this show about OLS?

```
plot(perform ~ eye)
eye_regr = lm(perform ~ eye)
abline(eye_regr, col="blue")
abline(h=mean(perform), col="red")
abline(v=mean(eye), col="green")
```

23. For the simple regression perform ~ skill, the code compares the model R-squared (about 0.046) with the squared correlation. Why does the script round the difference to 16 digits instead of testing with ==, and what equality is being checked?

```
skill_summary = summary(lm(perform ~ skill))
(skill_coef = cor(perform, skill)^2)
round(skill_summary$r.squared - skill_coef, 16)
```

24. The code fits perform on skill and ravens, and the last line of the summary reports F(2, 82) = 3.3 with p = 0.04184. State the null and alternative hypotheses of this F-test.

```
mult_lm = lm(formula = perform ~ skill + ravens)
(mult_summary = summary(mult_lm))
```

25. In the output F(2, 82) = 3.3 for the model perform ~ skill + ravens, what do the numerator degrees of freedom (2) and the denominator degrees of freedom (82) represent?

```
mult_summary = summary(lm(perform ~ skill + ravens))
mult_summary$fstatistic
```

26. The F-test of the model perform ~ skill + ravens gives F(2, 82) = 3.3 and p = 0.04184. At the 5% significance level, what should we conclude about the model as a whole?

```
# F(2,82) = 3.3, p = 0.04184 (alpha = 0.05)
```

27. The fitted equation of perform on skill and ravens is y = -4.355e-16 + 0.1712 * skill + 0.1746 * ravens, and all variables are standardized (mean 0, standard deviation 1). Interpret the coefficient 0.1712 and explain why the intercept is essentially 0.

```
mult_summary$coefficients
```

28. Read the code. It returns 0.3475 for a group whose skill is 0.5 and whose ravens score is 1.5. What does this number represent and how is it obtained from the model?

```
predict(mult_lm, data.frame(skill=0.5, ravens=1.5))
```

29. Read the code. It supplies two new cases, (skill 0.5, ravens 1.5) and (skill 1.5, ravens 0.5), and asks for a 95% confidence interval. What does the output contain for each case, and what population quantity does the interval describe?

```
predict(mult_lm, data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5)), interval =
"confidence", level = 0.95)
```

30. The code predicts over 10,000 x-values and shades the region between two columns of pred. What do pred[,2] and pred[,3] contain, and what does the shaded polygon show?

```
pred <- predict(eye_regr, data.frame(eye=points), interval = "confidence", level = 0.95)
polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col = rgb(0, 0, 0, 0.1), border = NA)
```

31. Suppose the call shown is changed from interval = "confidence" to interval = "prediction". How does the meaning and the width of the interval change?

```
predict(mult_lm, data.frame(skill=0.5, ravens=1.5), interval = "confidence")
```

32. Read the simulation. Each iteration draws a new sample and stores one number in estimator[rowIdx]. What is the number that is stored, and what does hist(estimator) display after 10,000 iterations (true intercept 3, true slope 0.2, sample size 10)?

```
for(rowIdx in 1:sampleCnt) {
  fertilizer = rnorm(sampleSize, 10, 2)
  yield = intercept + slope * fertilizer + rnorm(sampleSize, 0, 1)
  coef = summary(lm(yield ~ fertilizer))$coefficients
  estimator[rowIdx] = coef[2, 1]
}
hist(estimator)
```

33. In the simulation, t[rowIdx] = coef[2,3] and pValue[rowIdx] = coef[2,4] are stored. What do coef[2,3] and coef[2,4] contain?

```
coef = summary(lm(yield ~ fertilizer))$coefficients
t[rowIdx] = coef[2, 3]
pValue[rowIdx] = coef[2, 4]
```

34. Read the code. The true slope is 0.2, but the null hypothesis says the slope is 0. When is the null rejected, and what does the share of "Reject" in table(rejection) estimate? What would it estimate if the true slope were 0?

```
alpha = 0.05
rejection = pValue <= alpha
rejection = ifelse(rejection, "Reject", "Fail to Reject")
table(rejection)
```

35. Read the code. For 20 samples (true slope 2, sample size 12) it records inside. When is inside TRUE, and about how many of the 20 95% intervals do you expect to contain the true slope? Why might the count differ from that?

```
interval = confint(fer_regr)
lowerBound = interval["fertilizer", "2.5 %"]
upperBound = interval["fertilizer", "97.5 %"]
inside = lowerBound <= slope && slope <= upperBound
```

36. Read the code. How does changing the confidence level from the default 0.95 to 0.99 affect the width of the confidence intervals for the coefficients, and why?

```
confint(mult_lm)
confint(mult_lm, level = 0.99)
```

37. The R-squared of perform ~ skill + ravens is about 0.0745, while the single-predictor squared correlations are about 0.0459 (skill) and 0.0469 (ravens). The check round(mult_summary$r.squared - skill_coef, 2) == 0 returns FALSE. What does this say about adding ravens, and is R-squared the sum of the two single values?

```
mult_summary$r.squared
round(mult_summary$r.squared - skill_coef, 2) == 0
round(mult_summary$r.squared - ravens_coef, 2) == 0
```

38. Read the code from Lab2.R. Why is the first result always 0 apart from rounding error, why are the deviations squared in the second line, and why does the third line divide by n - 1?

```
sum(competence - mean(competence))
sum((competence - mean(competence))^2)
sum((competence - mean(competence))^2) / (length(competence) - 1)
```

39. For the variable grade, quantile(grade) gives a first quartile of 72 and a third quartile of 84. What does IQR(grade) compute, what do the box and whiskers of boxplot(grade) show, and why use the IQR when a distribution is skewed or has outliers?

```
quantile(grade)
IQR(grade)
boxplot(grade)
abline(h=median(grade), col="red")
```

40. For the practice data, the correlation of grade with competence is about 0.83 and with breaks about -0.85. Describe the general pattern each scatterplot shows.

```
plot(grade ~ competence)
plot(grade ~ breaks)
```

41. Read the function to12 from Lab4.R. What does to12("22") return and how? What happens for to12("12"), and why is that a flaw?

```
to12 <- function(str){
  hour = as.integer(str)
  if (hour > 12) {
    return(paste(hour - 12, "PM"))
  } else {
    return(paste(hour, "AM"))
  }
}
```

42. Read the code from Lab4.R. Why is sapply() used to convert the whole column, instead of calling to12(data$openingHour) directly?

```
(openingHour <- sapply(data$openingHour, to12))
(closingHour <- sapply(data$closingHour, to12))
```

43. Explain the role of the lm() function in R and how the formula y ~ x1 + x2 specifies the dependent variable and the independent variables.

```
model <- lm(y ~ x1 + x2, data = df)
summary(model)
```

44. Why is the as.factor() function used on a categorical variable such as session before running a regression in R?

```
model <- lm(y ~ x + as.factor(session), data = df)
```

45. Explain the difference between using "+", ":" and "*" between two variables x1 and x2 in an R model formula.

```
lm(y ~ x1 + x2)
lm(y ~ x1:x2)
lm(y ~ x1*x2)
```

46. What does a significant F-statistic of the overall regression, shown in the last line of summary(model), indicate about the relationship between the dependent variable and the independent variables?

```
summary(model)
```

47. How do you interpret the estimated coefficient of a continuous independent variable in a linear regression model?

```
summary(model)$coefficients
```

48. What is the purpose of the linearHypothesis() function from the car package in R?

```
library(car)
linearHypothesis(model, c("x1 = 0", "x2 = 0"))
```

49. In the regression lm(wage ~ female + educ), female is a dummy variable equal to 1 for women and 0 for men. How is the coefficient of female interpreted?

```
lm(wage ~ female + educ, data = df)
```

50. In the regression lm(wage ~ educ*female), what does the coefficient of the interaction term educ:female measure, and what does the coefficient of female alone measure?

```
lm(wage ~ educ*female, data = df)
```

51. The code compares a restricted model, with one regression for both groups, and an unrestricted model with a full set of interactions with the dummy female. What hypothesis does anova(restricted, unrestricted) test, and what assumption does this test make?

```
restricted <- lm(wage ~ educ + exper, data = df)
unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
anova(restricted, unrestricted)
```

52. In the code, the dependent variable inlf is binary (1 = in the labor force, 0 = not). What do the coefficients of this lm() regression measure, and what is a known drawback?

```
lpm <- lm(inlf ~ educ + age, data = df)
```

Econometrics R — Model answers

1. W1 · Variable assignment
`<-` and = are the **assignment** operators in R
The **right-hand side** (a value or the result of a calculation) is evaluated
The result is **stored** in the variable on the left-hand side
Key results: assign · right-hand side · stored · left-hand side

2. W1 · Vector creation
The function **`c()`** (concatenate) combines several values into **one vector**
`c(9, 8, 7, 6)` joins the four numbers into a single vector
Key results: c() · one vector · joins

3. W1 · Vector arithmetic
`try1 + 5` adds 5 to **each element** of the vector and gives 14 13 12 11
Two vectors of the same length are added **element by element**, position by position, so `try1 + try2` gives 14 12 10 8
Key results: each element · element by element

4. W1 · Structure of a data frame
Each **row** is one **observation** (a subject or a case)
Each **column** is one **variable**
`dim(df)` returns the number of rows and the number of columns
Key results: observation · variable

5. W1 · The summary() function
`summary()` reports the **minimum** and the **maximum**
It reports the **mean** and the **median**
It reports the first and third **quartiles**
For a data frame it does this for every column
Key results: minimum · mean · quartile

6. W1 · Mean versus median with outliers
The **median** is more appropriate
The mean is **sensitive** to outliers because every value enters the calculation, so one extreme value pulls it away from the center
The median depends only on the middle position of the sorted data, so it is hardly affected
Key results: median · sensitive

7. W1 · Reading a histogram
A histogram shows the **shape of the distribution**, for example skewness
It shows the **frequency** (count) of observations in each bin
It shows where the **center** of the data lies
`breaks` controls the number of bins
Key results: distribution · frequency · center

8. W1 · Finding the mode with table() and sort()
`table(x)` builds a **frequency table** of each value
`sort(table(x), decreasing = TRUE)` orders it in **descending** order
The value with the **highest frequency**, the first entry, is the mode
Key results: table() · sort() · highest frequency

9. W2 · Reading Description File and Extracting Names
The code reads the dataset description file and performs **string split** and **subsetting** operations to automatically **extract variable names** from WAGE2_description.txt
Key results: extract variable names · string split

10. W2 · Interpreting Log Wage in W2
Using **lwage** (the **natural log** of wage) linearizes multiplicative relationships and allows coefficients to be interpreted approximately as a **percentage change**
Key results: natural log · percentage change

11. W2 · Binary Dummy Variables in W2
Variables such as married and urban function as **binary dummy variables**, representing group membership whose coefficients measure the **difference in intercept** relative to the base category
Key results: binary dummy variables · difference in intercept

12. W2 · Handling Missing Values with na.strings
Setting **na.strings="."** automatically **replaces missing values** (represented by dots) with **NA**, which **prevents calculation errors** when computing summary statistics
Key results: replaces missing values · prevents calculation errors

13. W2 · Attaching and Detaching Data Frames
`attach()` allows **direct variable access** without typing the dataset prefix, but introduces the risk of a **name conflict** if identical variable names exist
Key results: direct variable access · name conflict

14. W2 · Sum of Mean Deviations Identity
The sum of deviations from the mean is always zero because **positive and negative deviations** around the mean perfectly **cancel out**
Key results: cancel out · positive and negative deviations

15. W2 · Variance vs Standard Deviation Units
Variance results in **squared units** that inflate error interpretation, whereas taking the square root reverts the metric back to the **original unit of measurement**
Key results: squared units · original unit of measurement

16. W2 · IQR Robustness under Skewness
The **IQR** measures the spread of the **middle 50 percent** of the data, making it **robust to outliers** that distort the mean and standard deviation
Key results: robust to outliers · middle 50 percent

17. W2 · Mosaic Formula Interface Advantages
The **mosaic package** utilizes **formula notation** (~), enabling clean **group-by calculation** without requiring repetitive manual subsetting or indexing
Key results: formula notation · group-by calculation

18. W2 · Package Namespace Resolution Operator
The double-colon operator performs **namespace resolution** to explicitly invoke a function, allowing you to **override mask** issues when package function names conflict
Key results: namespace resolution · override mask

19. W2 · Vectorized Function Application using sapply
Using **`sapply()`** achieves **vectorized application** of custom functions across vector elements, which **avoids explicit loops** and improves execution efficiency
Key results: vectorized application · avoids explicit loops

20. W2 · String Pattern Detection Logic
`str_detect()` executes **pattern matching** to evaluate whether a substring exists, returning a **boolean** **TRUE FALSE** value utilized in conditional logic
Key results: boolean TRUE FALSE · pattern matching

21. W5 · Scatterplot formula
In **`perform ~ eye`** the variable on the left, **perform**, is the dependent (outcome) variable and goes on the y-axis
the variable on the right, **eye**, is the independent (predictor) variable and goes on the x-axis
That is why the outcome must be written first
Key results: perform · eye

22. W5 · OLS line and the two means
The blue OLS line always **passes through** the point where the vertical line at the **mean of x** (eye) meets the horizontal line at the **mean of y** (perform), so the fitted line is anchored at the point of means
Key results: passes through · mean of x · mean of y

23. W5 · R-squared equals squared correlation
It checks that in a simple linear regression **R-squared equals the correlation squared**
The two numbers are computed by different formulas, so they can differ by **tiny floating-point (rounding) differences**
rounding the difference to 16 digits lets the check ignore that noise
Key results: r-squared equals correlation squared · small differences

24. W5 · Null and alternative of the F-test
The null hypothesis is that **all coefficients are 0** (the skill and ravens slopes are both 0, $\beta_1 = \beta_2 = 0$)
The alternative is that **at least one** of them is not 0
Key results: all coefficients are 0 · at least one

25. W5 · Degrees of freedom in the F-statistic
The numerator df, 2, is the **number of predictors** tested (skill and ravens)
The denominator df, 82, is the **residual degrees of freedom**, n - k - 1 = n - 3, which implies n = 85 observations
Key results: number of predictors · residual degrees of freedom

26. W5 · Conclusion of the F-test
Because the p-value 0.04184 is less than 0.05, we **reject the null hypothesis**
At least one of the coefficients is not 0, so the model explains perform **significantly better than a model with no predictors**
Key results: reject the null hypothesis · at least one · better than

27. W5 · Reading the coefficients
Holding ravens constant, a one **standard deviation** increase in skill is associated with a 0.1712 standard deviation increase in predicted perform
Because all variables are **standardized** (mean 0), the regression plane passes through the point of means, so the intercept is about 0
Key results: holding ravens constant · standard deviation · standardized

28. W5 · Meaning of predict()
The value is the **predicted value (point estimate)** of perform for skill = 0.5 and ravens = 1.5
It is obtained by **substituting** these values into the estimated **regression equation**: -4.355e-16 + 0.1712 * 0.5 + 0.1746 * 1.5, which is about 0.3475
Key results: predicted value · substitut

29. W5 · Columns returned with interval = "confidence"
For each case (one row per pair) the output gives **fit**, the predicted value (about 0.3475 and 0.3441), plus the **lower and upper** bounds (lwr, upr) of a 95% confidence interval
The interval expresses how accurate our estimate of the **population average** of perform is at those predictor values
Key results: fit · lwr · population average

30. W5 · Confidence band code
The columns of pred are fit, lwr and upr, so pred[,2] holds the **lower bounds** and pred[,3] holds the **upper bounds** of the 95% confidence interval at every x-value
The polygon fills the area between them, drawing a shaded **confidence band** around the regression line
Key results: lower bound · upper bound · confidence band

31. W5 · Confidence interval versus prediction interval
The confidence interval covers the **population average** of the dependent variable, whereas the prediction interval covers an **individual new observation** at those predictor values
Because an individual value also includes its own random error, the prediction interval is **wider**
Key results: individual · wider · population average

32. W5 · Monte Carlo loop
Every iteration draws a **new random sample**, regresses yield on fertilizer and stores the **slope estimate** (coef[2, 1], the Estimate of fertilizer)
The histogram of these 10,000 slopes shows the **sampling distribution** of the OLS slope estimator, centred near the true slope 0.2
Key results: slope estimate · sampling distribution · new random sample

33. W5 · Columns of the coefficient table
In the coefficient table the columns are Estimate, Std. Error, t value and `Pr(>|t|)`, and row 2 is the **slope** of fertilizer
So coef[2,3] is the **t-statistic** and coef[2,4] is the **p-value** of the slope test
Key results: t-statistic · p-value · slope

34. W5 · Rejection rule and what the table estimates
The null is rejected when the **p-value is less than or equal to alpha** (0.05)
With a true slope of 0.2 the null is false, so the share of Reject estimates the test's **power**
If the true slope were 0 (a true null), that share would estimate the **Type I error rate**, about alpha
Key results: p-value less than alpha · power · type I error

35. W5 · Coverage of confidence intervals
inside is TRUE when the **true slope lies within** that sample's interval [lowerBound, upperBound]
Across repeated samples about **95%**, roughly 19 of 20, of the intervals should contain it, but the count of any one run of 20 varies by **chance** (random sampling)
Key results: true slope inside · 95% · chance

36. W5 · Confidence level and interval width
The 99% intervals are **wider**
To be **more confident** that the interval captures the true parameter, the interval must be larger, which means a less precise estimate
Key results: wider · more confident

37. W5 · Comparing R-squared values
Adding ravens makes R-squared **larger** than either single-predictor value (the difference rounds to 0.03, so the check is FALSE)
But 0.0745 is smaller than 0.0459 + 0.0469, so R-squared is **not the sum**: the predictors **overlap** and share some of the explained variance
Key results: larger · not the sum

38. W5 · Deviations from the mean (Lab2.R)
Deviations from the mean always sum to **zero** because positive and negative deviations **cancel**
Squaring removes the signs so the deviations can be summed, and dividing by **n - 1** gives the **sample variance** (`var(competence)`)
its square root is `sd()`
Key results: always 0 · squar · n - 1

39. W5 · IQR and boxplot (Lab2.R)
`IQR(grade)` is the **third quartile minus the first quartile** (84 - 72 = 12), the range of the middle 50% of the data, drawn as the box
Whiskers extend to the most extreme points within **1.5 times** the IQR, and points beyond are plotted as **outliers**
The IQR is robust, unlike the mean and standard deviation, when data are skewed or have outliers
Key results: third quartile minus first quartile · 1.5 · outlier

40. W5 · Reading the scatterplots (Lab2.R)
Grade rises as **competence** rises, a **positive** relationship (r about 0.83)
Grade falls as the number of **breaks** rises, a **negative** relationship (r about -0.85)
Key results: positive · negative

41. W5 · Function logic (Lab4.R)
`to12("22")` returns **"10 PM"** because 22 is greater than 12, so the function **subtracts 12** and pastes "PM"
`to12("12")` returns **"12 AM"**: 12 is not greater than 12, so it falls in the else branch and **noon is mislabeled as AM**
Key results: 10 PM · subtract 12 · 12 AM

42. W5 · Why sapply (Lab4.R)
to12 uses **if**, which is not vectorized and handles only a **single value** at a time
`sapply()` applies the function to **each element** of the column and returns the results as a vector
Key results: each element · if

43. W6 · lm function syntax
The **`lm()`** function fits a **linear model** (an OLS regression)
In the **formula** `y ~ x1 + x2`, the tilde separates the **dependent variable** y on the left from the **independent variables** x1 and x2 on the right, which are joined by plus signs
Key results: linear model · dependent variable · independent variables · formula

44. W6 · Factor conversion and dummy variables
Using **`as.factor()`** tells R to treat the variable as a **categorical variable** rather than a number
`lm()` then **automatically** creates **dummy variables** for the categories and leaves one out as the **base (reference) category**
Key results: categorical variable · dummy variables · automatically · base category

45. W6 · Interaction terms in R formulas
"+" includes only the **main effects** of x1 and x2
":" includes only their **interaction** term (the product x1 * x2)
"*" is shorthand for x1 + x2 + x1:x2, so it includes **both** the main effects and the interaction
Key results: main effects · interaction · both

46. W6 · F-statistic interpretation
A significant F-statistic indicates the **overall significance** of the model
The null hypothesis that all slope coefficients are 0 is rejected, so **at least one** coefficient of the independent variables is **different from zero**
Key results: overall significance · at least one · not zero

47. W6 · Coefficients and economic significance
The coefficient measures the expected **marginal effect** on the dependent variable of a **one-unit change** in that independent variable, **holding** all other variables in the model **constant**
Key results: holding constant · unit change · marginal effect

48. W6 · Linear hypothesis testing
The **`linearHypothesis()`** function **tests** linear hypotheses, that is, linear **restrictions** on the **coefficients** of a fitted regression model, such as several coefficients being 0 at the same time or two coefficients being equal
For an lm object it reports an **F-test**
Key results: restrictions · test · coefficients · F-test

49. W6 · Coefficient of a dummy variable
The coefficient of female is the **difference** in the expected wage between women and the **base group** (men), **holding educ constant**
Key results: difference · base group · holding educ constant

50. W6 · Interaction with a dummy variable
The coefficient of `educ:female` is the **difference in slopes:** how much the effect of educ on wage (the **return to education**) for women differs from that for men
The coefficient of female alone shifts the **intercept**, the wage difference at educ = 0
Key results: difference in slopes · return to education · intercept

51. W6 · Testing differences across groups (Chow test)
The null hypothesis is that the **same regression** holds for both groups, that is, the coefficient of female and all its interaction coefficients are 0
The **F-test** compares the SSR of the restricted and unrestricted models (the Chow test), and it assumes a **constant error variance** across the groups
Key results: same regression · F-test · constant error variance

52. W6 · Binary dependent variable
This is a **linear probability model**
A coefficient measures the change in the **probability** that inlf = 1 for a one-unit change in the regressor
A drawback is that the fitted values can fall **outside 0 and 1**, for example a negative predicted probability
Key results: probability · linear probability model · outside 0 and 1